SERENA LI

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EDUCATION

University of Southern California Los Angeles, CA
Master of Science, Computer Science – Progressive Degree Program (GPA: 3.75) May 2027
Relevant Coursework: Machine Learning, Geospatial Data Analytics, Data Visualization, Analysis of Algorithms

University of Southern California Los Angeles, CA
Bachelor of Science, Computer Science – Progressive Degree Program (GPA: 3.88) May 2027
Relevant Coursework: Operating Systems, Internetworking, Artificial Intelligence, Human Computer Interaction, Data Structures
Involvements: HackSC Executive Director, Esri UC Student Assistant, Trojan Fencing Club Foil Fencer, Band Kori Acoustic Guitarist

EXPERIENCE

Optivide **Los Angeles, CA**
Software Engineer Fellow May 2026 – Aug 2026
- Built a cross-platform React Native app with a Google Cloud-hosted backend and database, connecting mobile workflows to cloud services.
- Applied specification-first prototyping with LLM tooling (Claude, ChatGPT) to rapidly translate product requirements into working application features.
- Architected automated code-review pipelines using GitHub Actions and Google Gemini API, streamlining PR validation and reducing feature implementation cycles by 50%.

ENXTI **Los Angeles, CA**
Founding Software Engineer Dec 2025 – Aug 2026
- Led a five-person team building React Native and Next.js applications in TypeScript, translating founder requirements into weekly MVP features, enabling soft-launch in 6 months.
- Engineered GraphQL integration with Shopify webhooks and developed MongoDB REST endpoints to synchronize real-time inventory for 5k+ product variants across 10 partner designers.
- Implemented user authentication using Firebase and integrated Stripe into the MVP, supporting account management and payment workflows.

Los Angeles Department of Transportation **Los Angeles, CA**
Software Engineer Intern Sep 2025 – Dec 2025
- Automated engineering workflows with reusable Python scripts and the Mapillary API, improving workflow speed by 20%.
- Fine-tuned YOLO object detection models on transportation imagery, achieving 90%+ confidence for automated municipal infrastructure analysis.

PROJECTS

WeatherDB - Climate Database Query Model **Los Angeles, CA**
Researcher Jun 2025 – Present
- Built WeatherDB's PyTorch pipeline as the sole technical contributor, creating a 19,500-query ERA5 benchmark to compare coordinate-based and polygon-vector representations.
- Automated Fourier encoding and multilayer perceptron (MLP) hyperparameter experiments with Optuna and CARC computing clusters, comparing configurations by validation MAE in studies of up to 200 completed trials.

Runway - Power Outage Forecast Web App *(github.com/Serena20003/see_your_power)* **Los Angeles, CA**
Lead Builder, Best Build with IBM Bob Winner at InnovateLA Dual-Use Hackathon Oct 2026
- Built a React and Python outage-preparedness prototype in 24 hours using IBM Bob Coding Agent, generating synthetic data to prototype forecasting despite labeled-data gaps.
- Integrated IBM's Watson AI platform into the chatbot and refined the React interface, connecting household questions to personalized preparation guidance.

MoneyTree - Investment Recommendation App *(moneytree-ah25.streamlit.app)* **Los Angeles, CA**
Lead Frontend Engineer, Best Use of Streamlit Winner at AthenaHacks Feb 2025
- Built MoneyTree's Streamlit interface to collect user preferences and display personalized investment recommendations, connecting matching and machine learning modules in one app.

SKILLS AND INTERESTS

Coding Languages: Python, TypeScript, C++, SQL, JavaScript
Frameworks & Libraries: React, React Native, PyTorch, Node.js, Next.js, Vue.js, Streamlit, Stripe API, Shopify API
Tools & Infrastructure: Git, GitHub Actions, GCP, AWS, MongoDB, Firebase, GraphQL, Docker
Languages: English (Native), Mandarin (Native), Teochew (Fluent), Shanghainese (Conversational), Cantonese (Conversational)